

PLamb – Summary Report - Barcelona

Executive Summary

This report captures the latest FR (Full Relativity) cosmological model run with $\Omega\Lambda$ fixed to zero (no dark energy). The analysis uses the Pantheon+SH0ES supernova dataset with full covariance and emcee sampling.

Run Configuration

Command used:

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6b.py' \  
--model FR \  
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \  
--Ol 0 --fix_Ol \  
--sampler emcee --nwalkers 32 --nsteps 4000 --nburn 1000 --  
verbose --show
```

Preliminary Best-Fit (short run, 16×100)

$H_0 \approx 68.07$, $\Omega_m \approx 0.371$, $\Omega\Lambda = 0$ (fixed), $A_{\text{acc}} \approx 0.107$, $n_{\text{acc}} \approx 0.529$, $\gamma_c \approx 0.293$, $\epsilon_M \approx -0.052$

$\chi^2/\text{dof} \approx 1.174$ (dof=1695), AIC ≈ 2001.5 , BIC ≈ 2034.1

Active parameters: H_0 , Ω_m , A_{acc} , n_{acc} , γ_c , ϵ_M ; Frozen: $\Omega\Lambda = 0$

Interpretation

The FR model without dark energy provides a statistically acceptable fit to Pantheon+SH0ES ($\chi^2/\text{dof} \approx 1.17$). This suggests that within FR, acceleration terms and small variations in $c(z)$ and $M(z)$ can compensate for the lack of $\Omega\Lambda$.

While this removes the need for dark energy explicitly, it introduces additional physics terms whose interpretation must be tested further against BAO and CMB.

Consistency & Next Steps

- Current output is from a short test run; production runs require longer chains (32×4000 steps with burn-in).
- Repeat runs are recommended to verify stability and convergence.
- Compare side-by-side with:

- 1) FR with $\Omega\Lambda$ free,
 - 2) Standard Λ CDM baseline.
- Save chains and compute autocorrelation times, ESS, and acceptance fractions to assess convergence.

Figures

FR | $H_0=68.07$, $\Omega_m=0.371$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=0.107$, $n_{\text{acc}}=0.529$, $\gamma_c=0.293$, $\epsilon_M=-0.052$
 $\chi^2/\text{dof}=1.174$ (k=6, AIC=2001.5, BIC=2034.1) | frozen: Ol=0

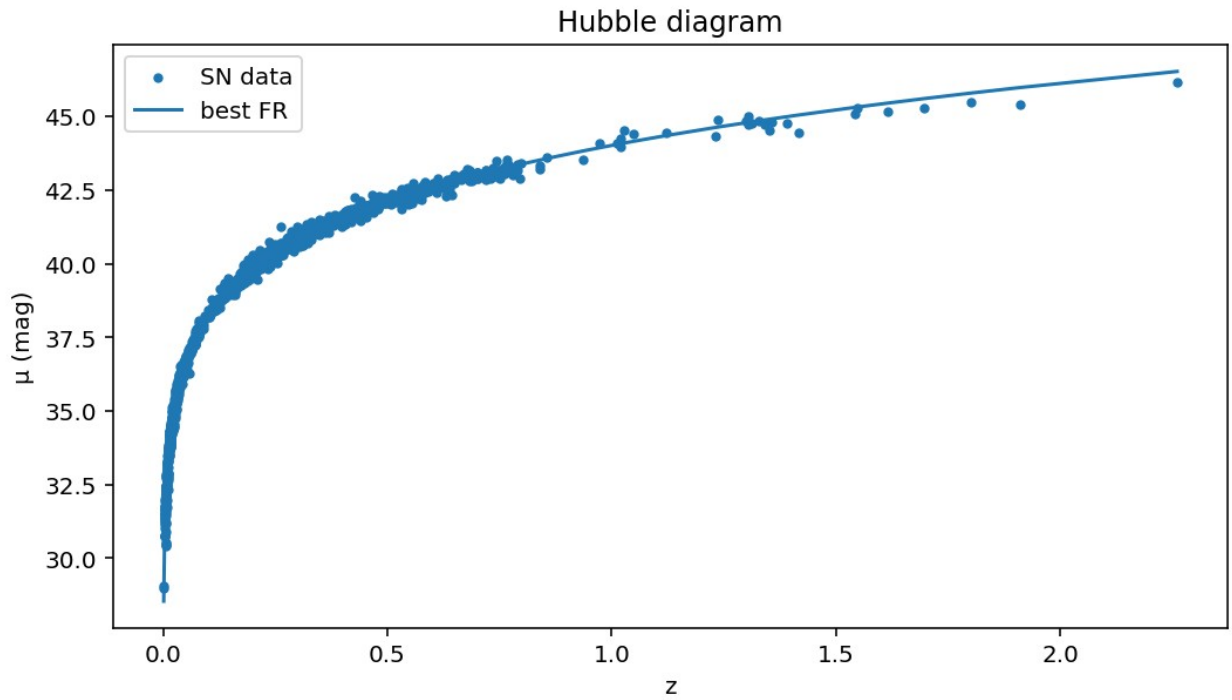


Figure: Hubble diagram, FR ($\Omega\Lambda=0$ fixed).

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